EASY · ALGEBRA · ANSWER KEY

Systems of two linear equations in two variables

03

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ANSWER KEY · SAT QUESTION BANK

2026-05-29

Q1 A

A. $4x + 2y = 86$
$3x + 5y = 166$

Choice A is correct. Hiro purchased 4 shirts and each shirt cost x dollars, so he spent a total of 4x dollars on shirts. Likewise, Hiro purchased 2 pairs of pants, and each pair of pants cost y dollars, so he spent a total of 2y dollars on pants. Therefore, the total amount that Hiro spent was 4x + 2y. Since Hiro spent $86 in total, this can be modeled by the equation 4x + 2y = 86. Using the same reasoning, Sofia bought 3 shirts at x dollars each and 5 pairs of pants at y dollars each, so she spent a total of 3x + 5y dollars on shirts and pants. Since Sofia spent $166 in total, this can be modeled by the equation 3x + 5y = 166.

Choice B is incorrect and may be the result of switching the number of shirts Sofia purchased with the number of pairs of pants Hiro purchased. Choice C is incorrect and may be the result of switching the total price each person paid. Choice D is incorrect and may be the result of switching the total price each person paid as well as switching the number of shirts Sofia purchased with the number of pairs of pants Hiro purchased.

Q2 B

B. $7, 19$

Choice B is correct. It's given by the second equation in the system that $y = 19$. Substituting 19 for y in the first equation yields $19 = 4x - 9$. Adding 9 to both sides of this equation yields $28 = 4x$. Dividing both sides of this equation by 4 yields $7 = x$. Therefore, since $x = 7$ and $y = 19$, the solution x, y to the given system of equations is $7, 19$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q3

A

A. 4, - 5

Choice A is correct. The solution to this system of linear equations is represented by the point that lies on both lines shown, or the point of intersection of the two lines. According to the graph, the point of intersection occurs when $x = 4$ and $y = -5$, or at the point 4, - 5. Therefore, the solution x, y to the system is 4, - 5.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q4

B

B. 15

Choice B is correct. Substituting 20 for $x+y$ in the second equation in the system yields $2(20)+3y=85$, or $40+3y=85$. Subtracting 40 from both sides of this equation yields $3y=45$. Dividing both sides of this equation by 3 yields $y=15$.

Choice A is incorrect. If $y=10$, then $x=10$ since $x+y=20$. However, substituting 10 for both x and y in the second equation yields $70=85$, which is a false statement. Choice C is incorrect. If $y=60$, then $x=-40$ since $x+y=20$. However, substituting these values for x and y in the second equation yields $220=85$, which is a false statement. Choice D is incorrect. If $y=65$, then $x=-45$ since $x+y=20$. However, substituting these values for x and y in the second equation yields $235=85$, which is a false statement.

Q5

A

A. $4, 28$

Choice A is correct. The second equation in the given system is $y = 28$. Substituting 28 for y in the first equation in the given system yields $28 = 12x - 20$. Adding 20 to both sides of this equation yields $48 = 12x$. Dividing both sides of this equation by 12 yields $4 = x$. Therefore, the solution x, y to the given system of equations is $4, 28$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the solution y, x, not x, y, to the given system of equations.

Choice D is incorrect and may result from conceptual or calculation errors.

Q6

D

D. 3

Choice D is correct. Dividing both sides of the equation $10x = 110$ by 10 yields $x = 11$. Substituting 11 for x in the equation $6x - 63 = y$ yields $6(11) - 63 = y$, which is equivalent to $66 - 63 = y$, or $3 = y$. Therefore, the value of y is 3.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the value of x, not y.

Choice C is incorrect and may result from conceptual or calculation errors.

Q7

A

A. $$\begin{aligned} s + p &= 250 \\ 5s + 12p &= 2{,}300 \end{aligned}$$

Choice A is correct. It's given that the petting zoo sells two types of tickets, standard and premium, and that s represents the number of standard tickets sold and p represents the number of premium tickets sold. It's also given that the petting zoo sold 250 tickets on one Saturday; thus, $s + p = 250$. It's also given that each standard ticket costs $5 and each premium ticket costs $12. Thus, the amount collected in ticket sales can be represented by $5s$ for standard tickets and $12p$ for premium tickets. On that Saturday the petting zoo collected a total of $2,300 from ticket sales; thus, $5s + 12p = 2{,}300$. These two equations are correctly represented in choice A.

Choice B is incorrect. The second equation in the system represents the cost per standard ticket as $12, not $5, and the cost per premium ticket as $5, not $12. Choices C and D are incorrect. The equations represent the total collected from standard and premium ticket sales as $250, not $2,300, and the total number of standard and premium tickets sold as $2,300, not $250. Additionally, the first equation in choice D represents the cost per standard ticket as $12, not $5, and the cost per premium ticket as $5, not $12.

Q8

A

A. 1

Choice A is correct. The first equation in the given system of equations is $x = 4$. Substituting 4 for x in the second equation in the given system of equations yields $y = 5 - 4$, or $y = 1$.

Choice B is incorrect. This is the value of x in the solution to the given system of equations, not the value of y.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q9 A

A. $(3,9)$

Choice A is correct. The solution to this system of linear equations is the point that lies on both lines graphed, or the point of intersection of the two lines. According to the graphs, the point of intersection occurs when $x = 3$ and $y = 9$, or at the point $(3,9)$.

Choices B and D are incorrect. Each of these points lies on one line, but not on both lines in the xy-plane. Choice C is incorrect. This point doesn't lie on either of the lines graphed in the xy-plane.

Q10 B

B. $3, 4$

Choice B is correct. If a point x, y lies on both lines in the graph of a system of two linear equations, the ordered pair x, y is a solution to the system. The graph shown is the graph of a system of two linear equations, where the two lines in the graph intersect at the point $3, 4$. Therefore, the point $3, 4$ lies on both lines, so the ordered pair $3, 4$ is the solution to the system.

Choice A is incorrect. The point $2, 3$ lies on one, not both, of the lines in the graph shown.

Choice C is incorrect. The point $4, 5$ lies on one, not both, of the lines in the graph shown.

Choice D is incorrect. The point $5, 6$ lies on one, not both, of the lines in the graph shown.

Notes

Accept equivalent forms (e.g. `0.25 ≡ 1/4`). For grid-in items, decimal and fraction answers are both valid.

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